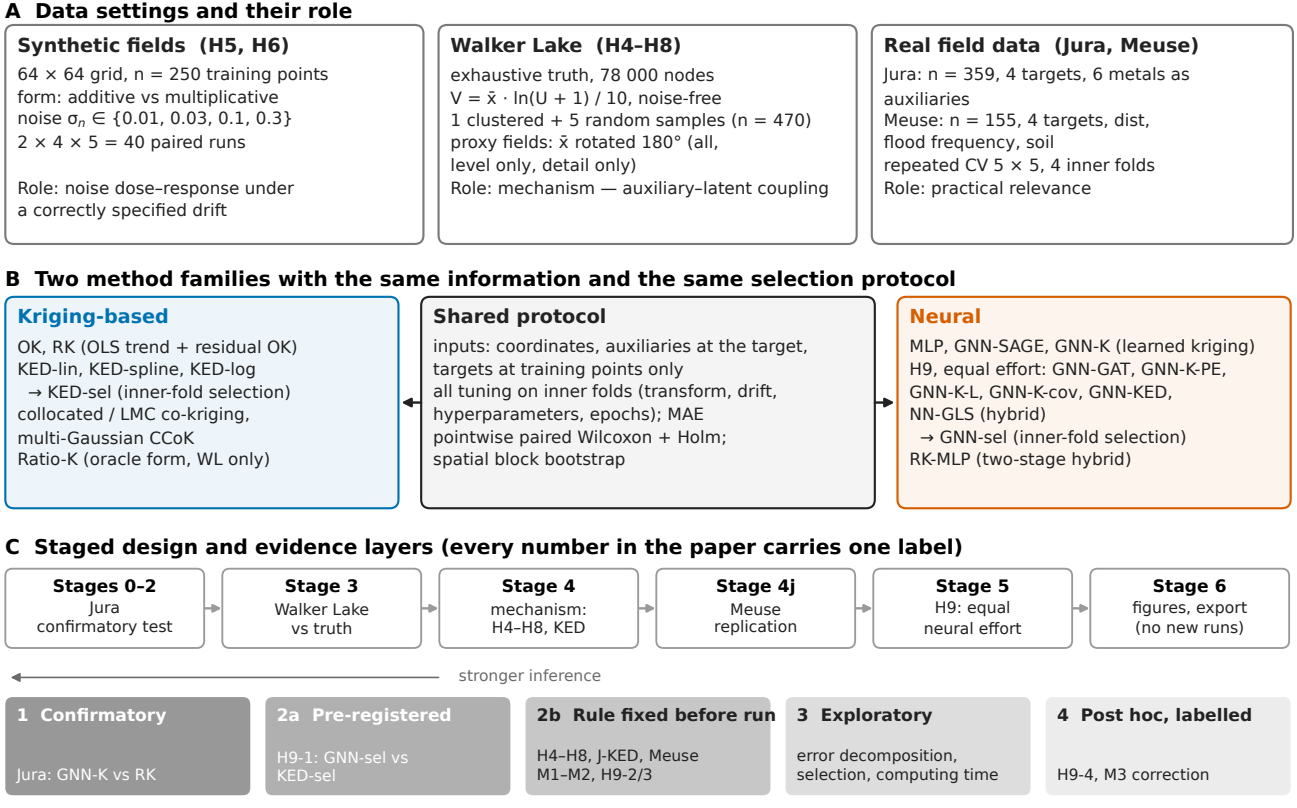
A Data settings and their role
Synthetic fields (H5, H6)
64 × 64 grid, n = 250 training points
form: additive vs multiplicative
noise $\sigma_n \in \{0.01, 0.03, 0.1, 0.3\}$
2 × 4 × 5 = 40 paired runs
Role: noise dose–response under a correctly specified drift
Walker Lake (H4–H8)
exhaustive truth, 78 000 nodes
$V = \bar{x} \cdot \ln(U + 1) / 10$, noise-free
1 clustered + 5 random samples (n = 470)
proxy fields: $\bar{x}$ rotated 180° (all, level only, detail only)
Role: mechanism — auxiliary–latent coupling
Real field data (Jura, Meuse)
Jura: n = 359, 4 targets, 6 metals as auxiliaries
Meuse: n = 155, 4 targets, dist, flood frequency, soil
repeated CV 5 × 5, 4 inner folds
Role: practical relevance
B Two method families with the same information and the same selection protocol
Kriging-based
OK, RK (OLS trend + residual OK)
KED-lin, KED-spline, KED-log
→ KED-sel (inner-fold selection)
collocated / LMC co-kriging,
multi-Gaussian CCoK
Ratio-K (oracle form, WL only)
Shared protocol
inputs: coordinates, auxiliaries at the target,
targets at training points only
all tuning on inner folds (transform, drift, hyperparameters, epochs); MAE
pointwise paired Wilcoxon + Holm;
spatial block bootstrap
Neural
MLP, GNN-SAGE, GNN-K (learned kriging)
H9, equal effort: GNN-GAT, GNN-K-PE, GNN-K-L, GNN-K-cov, GNN-KED,
NN-GLS (hybrid)
→ GNN-sel (inner-fold selection)
RK-MLP (two-stage hybrid)
C Staged design and evidence layers (every number in the paper carries one label)
Stages 0–2
Jura
confirmatory test
Stage 3
Walker Lake
vs truth
Stage 4
mechanism:
H4–H8, KED
Stage 4j
Meuse
replication
Stage 5
H9: equal
neural effort
Stage 6
figures, export
(no new runs)
stronger inference
1 Confirmatory
Jura: GNN-K vs RK
2a Pre-registered
H9-1: GNN-sel vs KED-sel
2b Rule fixed before run
H4–H8, J-KED, Meuse
M1–M2, H9-2/3
3 Exploratory
error decomposition,
selection, computing time
4 Post hoc, labelled
H9-4, M3 correction